

Series

Def. From a sequence

$$\{x_n\}_{n=1}^{\infty} = x_1, x_2, x_3, \dots$$

we can form the series

$$\sum_{n=1}^{\infty} x_n = x_1 + x_2 + x_3 + \dots$$

σ , Σ sigma

Can such an Infinite process/sum be completed? Need a careful precise mathematical def of what $\sum_{n=1}^{\infty} x_n$ actually means.

Partial sums of a series are finite ~~initial~~ initial sums

$$S'_k = \sum_{n=1}^k x_n = x_1 + x_2 + \dots + x_k.$$

Def 2.2.1 Convergence for series.

$$\sum_{n=1}^{\infty} x_n := \lim_{k \rightarrow \infty} \left(\sum_{n=1}^k x_n \right)$$

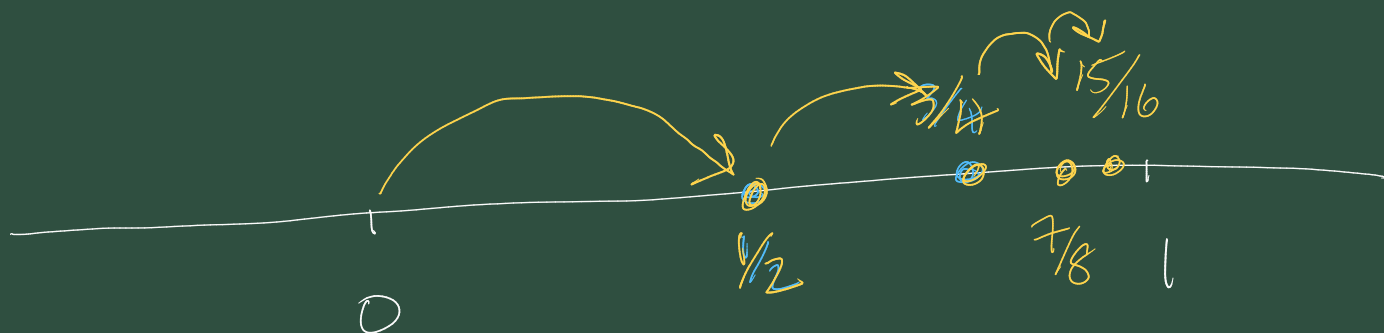
Ex 2.2.2

$$\sum_{m=1}^{\infty} m = 1 + 2 + 3 + 4 + \dots$$

Divergent, increasing and unbounded.

Ex 2.2.3 A convergent example.

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$



At each stage we advance towards 1 by half the remaining distance.

$$\text{So } \sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = 1.$$

$$\text{as } S_k = \frac{2^k - 1}{2^k} = 1 - \frac{1}{2^k}$$

$\rightarrow 1$ as $k \rightarrow \infty$

by algebra of limits theorem

$$\text{as } \frac{1}{2^k} \rightarrow 0$$

We can observe that $\sum_{m=1}^{\infty} m$ diverges as the terms, m , are too big.

and that $\sum_{m=1}^{\infty} (\frac{1}{2})^m$ converges as the terms, $\frac{1}{2^m}$, are small enough.

But just because the series terms get smaller and smaller, does not guarantee/imply convergence.

Eg 2.2.4 Harmonic series.

$$\sum_{m=1}^{\infty} \frac{1}{m} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

Consider a suitable grouping of these terms

$$\begin{aligned} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ &\quad + \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \left(\frac{1}{17} + \dots + \frac{1}{32}\right) \\ &\quad + \dots \end{aligned}$$

, grouping 2^k terms together for k^{th} group $k=1, 2, \dots$

$$\begin{aligned}
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) \\
&\quad + \left(\frac{1}{16} + \dots + \frac{1}{16}\right) + \left(\frac{1}{32} + \dots + \frac{1}{32}\right) \\
&\quad + \dots \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \\
&\quad + \frac{1}{2} + \frac{1}{2} \\
&\quad + \dots
\end{aligned}$$

this increases without upper bound and so is divergent. and so the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ also diverges.

An "algebra of limits" type theorem for series.

We can do this for linear combinations.

Th. Assuming $\sum_{m=1}^{\infty} a_m = A$
and $\sum_{m=1}^{\infty} b_m = B$ then for any

$\alpha, \beta \in \mathbb{R}$.

$$\sum_{m=1}^{\infty} (\alpha a_m + \beta b_m) = \alpha A + \beta B.$$

Proof (An application of the def.
of $\sum$, and algebra of limits
theorem for sequences).

$$\alpha A + \beta B = \alpha \left(\sum_{m=1}^{\infty} a_m \right) + \beta \left(\sum_{m=1}^{\infty} b_m \right)$$

$$= \alpha \lim_{k \rightarrow \infty} \left(\sum_{m=1}^k a_m \right) + \beta \lim_{k \rightarrow \infty} \left(\sum_{m=1}^k b_m \right)$$

$$= \lim_{k \rightarrow \infty} \left(\alpha \left(\sum_{m=1}^k a_m \right) + \beta \left(\sum_{m=1}^k b_m \right) \right)$$

, by algebra of limits for
seos.

$$= \lim_{k \rightarrow \infty} \left(\sum_{m=1}^k (\alpha a_m + \beta b_m) \right)$$

$$= \sum_{m=1}^{\infty} (\alpha a_m + \beta b_m), \text{ by the def. of } \sum_{m=1}^{\infty}$$

Convergence Tests

Theorem General Term Test.

This can recognize divergence.

Claim. $\sum_{m=1}^{\infty} a_m$ converges $\Rightarrow \lim_{m \rightarrow \infty} a_m = 0$

Proof Use formal ε, N def of sequence convergence.

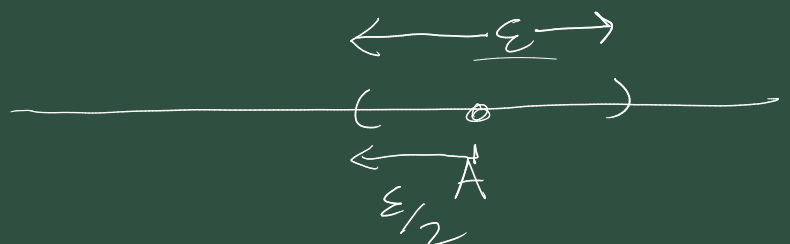
Assume $\sum_{m=1}^{\infty} a_m = A$, for some $A \in \mathbb{R}$.

i.e. $A_k = \sum_{m=1}^k a_m \rightarrow A$ as $k \rightarrow \infty$

So given any $\varepsilon > 0$ there exists N such that $\forall k \in \mathbb{Z}^+$

$$\forall k \geq N \quad |A_k - A| < \varepsilon/2$$

(since $A_k \rightarrow A$ as $k \rightarrow \infty$).



$$\Rightarrow \forall k \geq K \quad |A_k - A_{k+1}| < \varepsilon$$

$$\Rightarrow \forall k \geq K \quad |-a_{k+1}| < \varepsilon$$

$$\Rightarrow \boxed{\forall k \geq K \quad |a_{k+1} - 0| < \varepsilon.}$$

So $\lim_{n \rightarrow \infty} a_n = 0.$

Th. Geometric Series Test.

For $a \neq 0$, ~~and~~

$\sum_{n=1}^{\infty} ar^{n-1}$ converges iff $|r| < 1$

in which case $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$

Proof If $a=0$

$$\sum_{n=1}^{\infty} ar^{n-1} = 0 + 0 + 0 + \dots = 0$$

Assume $a \neq 0$.

If $r=1$

$$\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=1}^{\infty} a = a + a + a + \dots$$

which clearly diverges.

Now assume $a \neq 0, r \neq 1$

Consider

$$S_k = \sum_{n=1}^k ar^{n-1} = a + \cancel{ar} + \cancel{ar^2} + \dots + \cancel{ar^{k-1}}$$
$$rS_k = \quad = \quad \cancel{ar} + \cancel{ar^2} + \dots + \cancel{ar^{k-1}} + ar^k$$

$$S_k - rS_k = a - ar^k$$

$$\Rightarrow S_k = \frac{a(1-r^k)}{1-r}, \quad \text{since } r \neq 1$$

By the algebra of limits for sequences

$$\lim_{k \rightarrow \infty} S_k = \frac{a(1 - \lim_{k \rightarrow \infty} r^k)}{1-r}$$

Notice that for $r = -1$ or $|r| > 1$

r^k is divergent, so S_k will diverge also and $\sum_{n=1}^{\infty} ar^{n-1}$ will be divergent.

But when $\lim_{n \rightarrow \infty} |r| < 1$, $r^k \rightarrow 0$ as $k \rightarrow \infty$.
So in this case we get

$\lim_{k \rightarrow \infty} S_k = \frac{a}{1-r}$, so the
 geometric series converges and
 $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$ (provided
 $a \neq 0, |r| < 1$)

Some examples

Consider again $\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m$

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = \sum_{m=1}^{\infty} \underbrace{\frac{1}{2}}_a \underbrace{\left(\frac{1}{2}\right)^{m-1}}_{r^{m-1}}$$

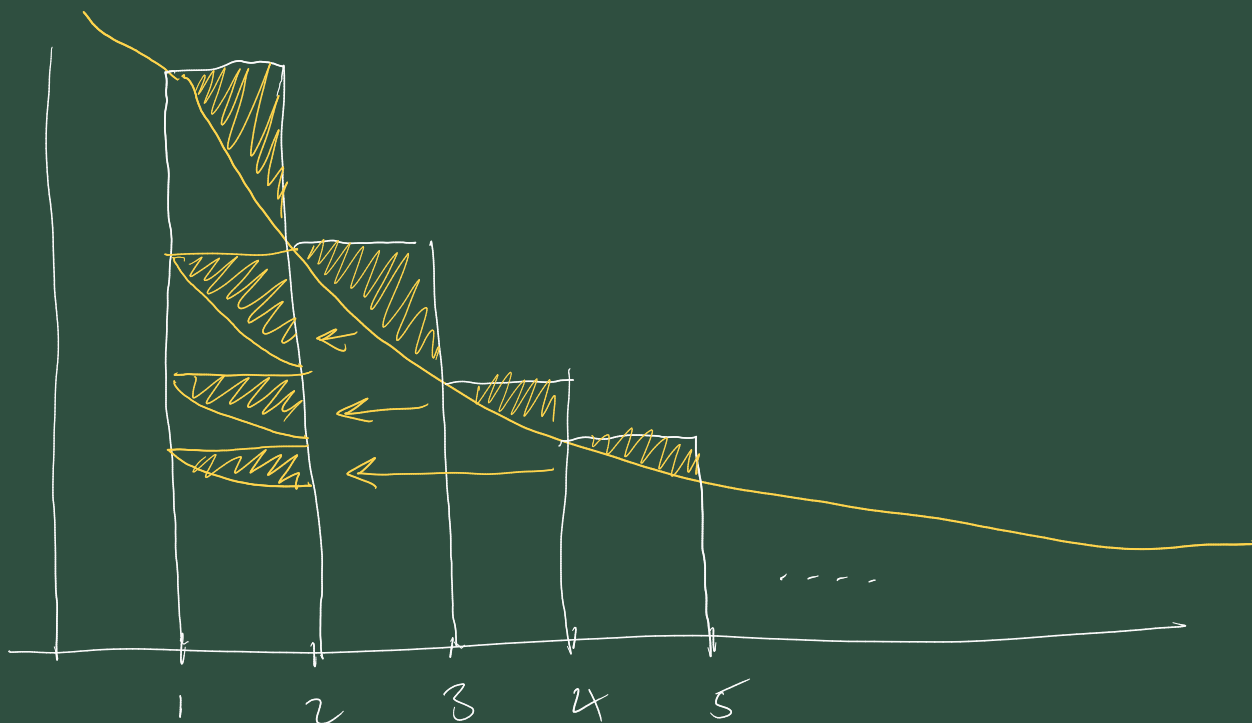
$r = \frac{1}{2}$ and $|\frac{1}{2}| < 1$. So this converges
 with sum

$$\sum_{m=1}^{\infty} \frac{1}{2} \left(\frac{1}{2}\right)^{m-1} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1.$$

Consider now $\sum_{n=1}^{\infty} x_n$

where

$$\begin{aligned}
 x_n &= \frac{2}{3} \left(\frac{1}{2}\right)^n + \frac{1}{4} \left(\frac{2}{3}\right)^n \\
 &= \frac{1}{3} \left(\frac{1}{2}\right)^{n-1} + \frac{1}{6} \left(\frac{2}{3}\right)^{n-1}
 \end{aligned}$$



$$\sum_{m=1}^{\infty} a_m = \sum_{m=1}^{\infty} f(m) = \text{sum of areas of rectangles.}$$

$$= \int_1^{\infty} f(t) dt + \sum \text{errors}$$

Consider the sum of the errors
By the assumptions that f is decreasing and always positive, these errors all fit within first rectangle, without any overlaps.

So we can say $\sum_{m=1}^{\infty} \text{errors}_m$ converges to a finite sum $\gamma \in \mathbb{R}$.

$$\text{So } \sum_{m=1}^{\infty} a_m = \int_1^{\infty} f(t) dt + \gamma.$$

$\sum_{n=1}^{\infty} a_n$ converges $\iff \int_1^{\infty} f(t) dt$ does.

Can apply this to harmonic series

$$\sum_{n=1}^{\infty} 1/n$$

$$\int_1^{\infty} \frac{1}{t} dt = \lim_{T \rightarrow \infty} \int_1^T \frac{1}{t} dt.$$

$$= \lim_{T \rightarrow \infty} [\ln(t)]_1^T$$

$$= \lim_{T \rightarrow \infty} \ln(T)$$

~~is~~ This is divergent, increases without upper bound.

$$\Rightarrow \sum_{n=1}^{\infty} 1/n \text{ is divergent.}$$